Quantum Polyhedra: An Addendum

July 24, 2022

Following my last paper, I have decided to correct some rushed thoughts, and introduce some new more concrete ideas through modern methods and tools.

Having spent time listening to the late Professor Penrose (and Rovelli, amongst others) in interview attempting to explain the theory of spin networks (a central part of Loop Quantum Gravity), as well as its offshoots, I have come to realise that either I am very stupid (possible!), or his theory doesn't pass the "Explain it to a High-School-Er" test.

For this reason, I am going to give up trying to understand, and (mostly) move on.

What I plan to do is take some of the nice ideas from the previous research paper, and run with them in other contexts.

For reference, the previous paper can be seen at these addresses:

`https://stackedflows.github.io/Main-Site/research.html`

`https://www.researchgate.net/publication/338712463_Quantum_Polyhedra`

1 Outline of paper

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2 Inaccuracies from the Prior Paper

Some sections that I know are rushed, but still have merit and don't need to be completely embarrassed are the following:

$$\left[0.4, 0.5, 0.6, 0.7, 0.8\right] \tag{1}$$

The saving factor, with these sections, is they all have a huge amount of literature where computations are done by hand and explained in nice detail.

It's my feeling that the following sections of the previous paper warrant a "rushed, pay no attention" certificate. By that I mean it's entirely possible that they are nonsense.

$$\left[0.9, 0.10, 0.11, 0.12, 0.13\right] \tag{2}$$

The main reason for this (I imagine) was a lack of exposure to the correct computational tools to model the systems that I was researching, at the level of computational expertise that I was at (a fancy way to say I was ignorant, and probably still am ignorant).

However, it has come to my attention that there is work occurring in this area that is likely worth keeping an eye on.

Before I introduce that research, I will delve a bit into the problem at hand.

3 Re-framing the Problem

Where in the previous paper did it all fall apart? Where did the "High-School-Er" explanation fault-er? Let's start simple.

3.1 Polyhedrons

Definition 3.1. *Polyhedron*

In geometry, a polyhedron is simply a three-dimensional solid which consists of a collection of polygons, usually joined at their edges.

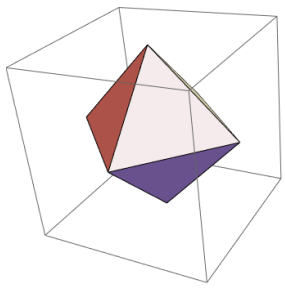

Figure 1: A simple Octahedron (coupled tetrahedron)

3.2 Graphs (Networks)

Definition 3.2. *Graph*

A graph, or network, is a group of nodes, interconnected with edges.

Figure 2: `https://www.wolframphysics.org/universes/wm1116/`

3.3 Polyhedral Networks (a comment on Spin Networks)

So naturally, we ask: is this what space is? Some sort of Network of Polyhedrons? What does that even look like?

3.3.1 Initial Randomness?

Maybe space is inherently random and overlapping with a lack of structure, and in some situations it seems to find some geometric equilibrium: a sort of quantum analogue to:

`https://www.youtube.com/watch?v=GKvT1lRWhE0&ab_channel=BBC`.

The bubbles go from "Jumbled" to "Convexified".

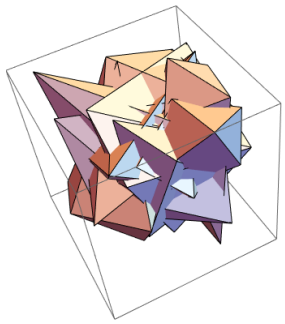

Figure 3: Jumbled Geometries

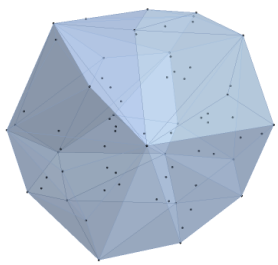

Figure 4: "Convexified" Geometries

The the above is constructed at:

`https://github.com/stackedflows/wolfram/blob/main/convexify.pdf`

3.3.2 Initial Order?

Perhaps it's the other way round: there is no such thing as non-tessellated space, and when we look at any location it is some variation on the following geometry.

```
TriangulateMesh[
MeshRegion[
{{0,0,0},{1,0,0},{1,1,0},{0,1,0},{0,0,1},{1,0,1},{1,1,1},{0,1,1}},
Hexahedron[{1,2,3,4,5,6,7,8}]],
MeshCellHighlight->{{2,All}->Opacity[0.5,Orange]}]
```

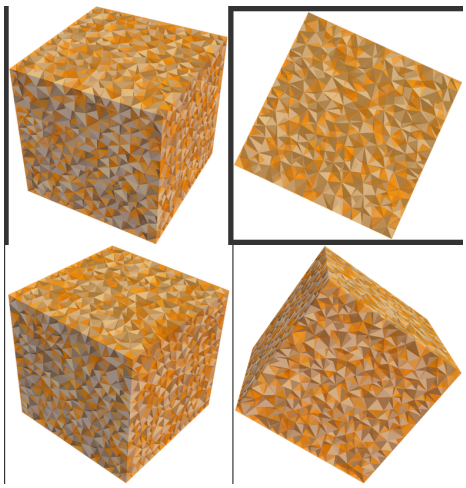

Figure 5: Tessellated Tetrahedrons within a cube of observed space

The software for a programmatic way to explore this sort of space is something I will be looking into!

3.4 The Problem

As you see, when we start asking questions about the evolution of interacting geometries, and you need to do massive calculations on matrices to get your point across, suddenly peoples eyes glaze over: things start getting hairy, simplicity disappears.

So whats next? Large volumes of calculations: then make visualisations that can be interpreted by humans easily.

The following papers provide some deeper insight into this possibility, using Wolfram Language Graph visualisations for LQG:

`https://community.wolfram.com/groups/-/m/t/2311732`

`https://community.wolfram.com/groups/-/m/t/2311181`

4 Conclusions and Further Work

Essentially, the Wolfram Physics project is in the process of making the world of spin networks human compute-able.

What does this mean? Well, for starters a programmatic analysis of the Quantum Tetrahedron and Coupled Tetrahedron will become trivial, as inbuilt Mathematica functions will do it all as soon as the "Loop Quantum Gravity" module is complete.

So then the hope is that people will decide do do a huge number of calculations with the LQG model in Mathematica. Or take inspiration from the Physics Project and design their own software to do a more specific version of the same thing, and then an intuition for the validity of LQG will really be within grasp of many more people.

It's worth noting that for the Wolfram guys, LQG is a "side project", as ridiculous as that sounds. The main event is in fact the Fundamental Physics project.

All of this, in addition to the sudden development in interest and adoption of automated proof engines such as the LEAN Language:

`https://leanprover.github.io/`,

and it seems that the future of Mathematics has never been so interesting.

If the reader was not already convinced, I hope this prompted them to get good with Mathematica.